Fiscal Year:	FY 2009	**Task Last Updated:**	FY 05/14/2010
PI Name:	Russo, Dane Ph.D.		
Project Title:	Man-Systems Integration Standards (MSIS) Reinvention		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2006	**End Date:**	09/30/2009
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Dane Russo, PhD, has taken over as PI (11/2006) from David Fitts. See under Fitts for previous task project.		
COI Name (Institution):	Watson, Andrew (NASA Ames Research Center)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Prior to 1987, NASA documentation included a number of different field center human engineering standards developed to help ensure proper definition and implementation of human system interface requirements for its human spaceflight programs. The Man-Systems Integration Standards (MSIS), NASA-STD-3000, first published in 1987, was intended to replace the field center standards with a single Agency-level human system standard. NASA-STD-3000 was revised and updated in 1995 and was the culmination of many months of effort by a large group of representatives from NASA, the prime aerospace contractors, subcontractors and other Government agencies. The final product was a comprehensive multivolume set containing integration design considerations, design requirements, and example design solutions. The NASA-STD-3000, though very comprehensive and up to date when it was published, is in need of revision and update if the standard is to meet its intended purpose for future human spaceflight programs. This is an appropriate time		

	for that revision since the JSC Space Life Sciences Directorate, under direction from the NASA Office of the Chief Medical Officer (OCHMO), is developing a single set of standards that cover all aspects of crew health, safety, and performance. These new standards will serve as an anchor point by which to judge compliance with acceptable levels of risk in the “Standards to Deliverables” approach to mitigation of the crew health, safety and performance risks associated with human spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These standards will serve as an anchor point by which to judge compliance with acceptable levels of risk in the “Standards to Deliverables” approach to mitigation of the crew health, safety and performance risks associated with human spaceflight.
Task Progress:	To address the problems described above, a project team created new standards and an accompanying reference design handbook. The team started with NASA-STD-3000 and expanded and updated the information using subject matter experts both within and outside NASA. New topics include address crew coordination, human cognition, and cognitive workload. The standard is called NASA Space Flight Human System Standard (SFHSS) and it consists of two-volumes: Volume 1 – Crew Health. This volume covers standards needed to support astronaut health (medical care, nutrition, sleep, exercise, etc.). This volume was completed through the funding and expertise within the Space Medicine Office of the Space Life Sciences Directorate at Johnson Space Center. Volume 2 – Human Factors, Habitability, and Environmental Health. This volume covers the standards for system design that will maintain astronaut performance (for example, environmental factors, design of facilities, layout of workstations, and lighting requirements). It includes classic human-factors requirements. This volume was completed by the Habitability and Human Factors Office of the Space Life Sciences Directorate at Johnson Space Center. It was funded by the Human Research Program. A reference design handbook, the Human Integration Design Handbook (HIDH) accompanies NASA-STD-3001. Much of NASA-STD-3001 is written in general terms so that it is applicable to the broad range of present and future NASA systems and missions. Many of the requirements are not verifiable, but NASA-STD-3001 (Volume 2) states that all new programs shall prepare verifiable system-specific requirements that will meet the general standards. For example, the document does not specify a design should accommodate a specific percentile range of a defined population. Rather, Volume 2 states that programs shall prepare program-specific requirements that define the user population and their size ranges. The design shall then accommodate the full size range of those users. NASA-STD-3001 does contain specific, verifiable requirements when NASA is sure the requirement will carry across all space systems and environments. The Human Integration Design Handbook accompanies the standard. It is approximately 1200 pages and covers the same topic areas as the standard. However, it does not contain specific requirements. For each topic, it provides background technical information, lessons learned, research needs, and references. HIDH serves two major purposes: 1. HIDH is a resource for writing the human-factors requirements for specific systems. 2. HIDH contains design guidance information to help ensure designers create a system that safely and effectively accommodates the capabilities and limitations of spaceflight crews. HIDH is a NASA Handbook, not a standards document. Advantages of the handbook are the flexibility in format and the relative ease of updating. It is meant to keep current with research findings. Initially the handbook and standard will be posted on the web in Adobe Acrobat format. The handbook can now be accessed at http://ston.jsc.nasa.gov/ . Because the handbook is so large and detailed, plans are to eventually put the handbook within a program that will facilitate search, feedback, and creation of personalized condensed versions. A Standard-Handbook Development Team, within the Habitability and Environmental Factors Division (HEFD) has the overall responsibility for maintaining, updating, and managing the Standard and Handbook. The Team consists of individuals with an overall knowledge of the Standard and Handbook. The Team will be responsible for making corrections as they are found and updating the documents as appropriate to advances in our knowledge and technology. The Standard is a NASA Agency document that is owned by the Office of the Chief Medical Officer and managed by the HEFD, Johnson Space Center, through the NASA Engineering Standards Program (NESP), Marshall Space Flight Center. Per NPD 8070.6, Technical Standards, NASA standards are updated every 5 years. The Handbook is also managed by the HEFD in conjunction with the JSC Scientific and Technical Information Program, which is a part of the JSC Information Resource Directorate. The Handbook will be updated at least once every two years.
Bibliography Type:	Description: (Last Updated: 05/18/2010)
NASA Technical Documents	NASA Office of the Chief Health and Medical Officer. "NASA Space Flight Human System Standard, Volume 2, Human Factors Habitability and Environmental Health." Houston, Tex.: NASA Lyndon B. Johnson Space Center, 2010. NASA-STD-3001. , Jan-2010
NASA Technical Documents	NASA Office of the Chief Health and Medical Officer. "Human Integration Design Handbook." Houston, Tex.: NASA Lyndon B. Johnson Space Center, 2010. NASA/SP-2010-3407. , Jul-2010